Graphing a System of Inequalities

Recall from Unit 3 how to graph the solution of a system of inequalities.

Example

Hide

Graph: $\begin{cases} y < -x - 1 \\ y \leq 4x - 3 \end{cases}$ Rewrite as equations:

slope = -1, y-int = -1: $y = -x - 1$ (1)

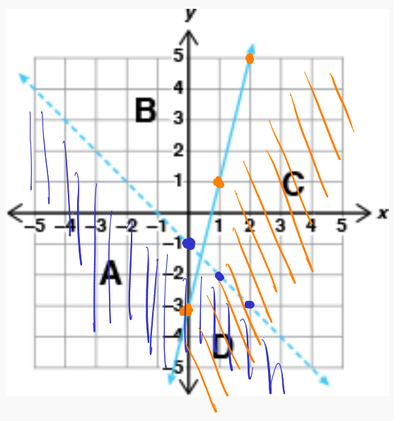
$y = 4x - 3$ (2) slope = 4, y-int = -3

Step 1

Graph both lines on a coordinate plane.

- Use a dashed line if the inequality contains $>$ or $<$.
- Use a solid line if the inequality contains $\geq$ or $\leq$.

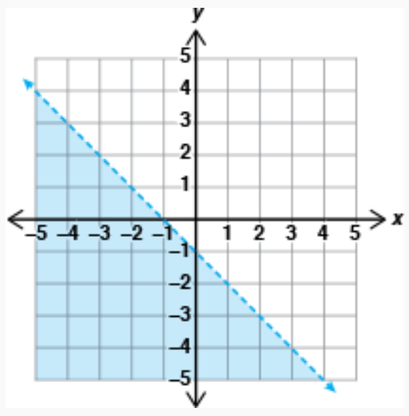
The two lines intersect to create four regions. Here they are labeled A, B, C, and D. One of these regions is the solution to the system.



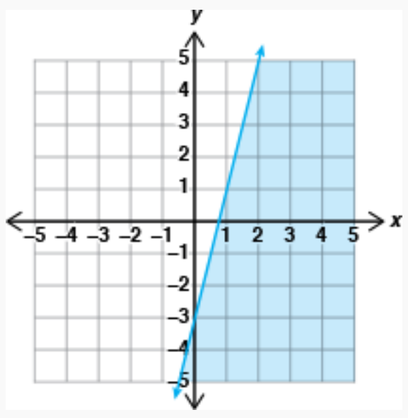
Step 2

Figure out which region is shaded for each inequality. The solution is the space or spaces where the solutions to the individual inequalities overlap.

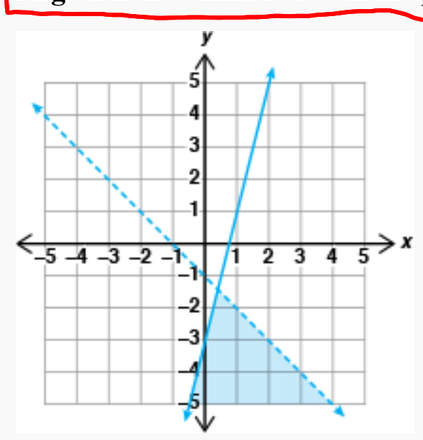
The graph of $y < -x - 1$ includes regions A and D.



The graph of $y \leq 4x - 3$ includes regions C and D.



Region D is common to both inequalities. Any point in this region makes both inequalities true.



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Lesson 5: Graphing Systems of Equations and Inequalities

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